Storm Sequence Control

User Guide — Parameter Reference and Pipeline Narrative

MKM Research Labs

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MKM Research Labs

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Contents

1 Introduction

The MKM Physical Risk platform generates a catalogue of synthetic storm sequences to quantify flood exposure across a property portfolio. Rather than relying on a handful of historical events, the system constructs **20,000 stochastic storm sequences**, each representing a plausible 168-hour (seven-day) weather event. These sequences drive a physics-based pipeline that propagates rainfall through river gauges and onto individual properties, producing flood depths and financial losses.

The **Storm Sequence Control** screen centralises all parameters that govern this pipeline. Every parameter has a default value calibrated to the Thames catchment, but operators may adjust them to explore sensitivities, stress-test assumptions, or adapt the model to a different catchment.

This guide explains the pipeline as a narrative—what each stage achieves physically, which parameters control it, and what happens when those parameters are moved up or down. Chapter 8 provides a condensed reference table for every parameter.

2 Pipeline Overview

The storm stress pipeline operates in five stages, each controlled by its own parameter section on the Control screen.

1. **Storm Generation** — Build 20,000 synthetic storm sequences, each containing one or more precipitation pulses fitted into a 168-hour event window.
2. **Hydrograph Synthesis** — Convert precipitation pulses into gauge-level water-surface-elevation time series using a gamma-shaped template, antecedent saturation, superposition, and infiltration.
3. **Gauge-to-Property Propagation** — Translate gauge readings into property-level flood depths using Manning's equation, exponential retention decay, and terrain-velocity scaling.
4. **Spatial Correlation** — Distribute catchment-mean rainfall unevenly across gauges via an exponential correlation kernel, so that nearby gauges receive similar rainfall while distant gauges differ.
5. **Stress Catalogue** — Filter the generated sequences to identify those that breach alert thresholds, forming the stress testing catalogue used by the trading desk.

Each stage reads its parameters from a single JSON file (`storm_control.json`) which the Control screen edits. Changes take effect immediately—no code deployment is required.

3 Storm Generation

Storm generation is the first and most complex stage. It answers the question: *what does the precipitation look like during this 168-hour window?*

3.1 The 168-Hour Event Window

Every storm sequence occupies exactly 168 hours (seven days). Within this window, one or more precipitation pulses occur, separated by drainage gaps. Two hard constraints apply:

- All precipitation must **end by hour 156**, leaving at least 12 hours of guaranteed drainage (the `min_drainage_window_hours` parameter).
- The total duration of all pulses plus all gaps must **fit within the budget** of 156 hours.

Increasing the event window allows longer storm sequences and more drainage time. Decreasing the minimum drainage window permits precipitation closer to the end of the window, reducing the guaranteed recession tail.

3.2 Batch Generator: Intensity Category Weights

The batch generator decides how many of the 20,000 sequences fall into each intensity category. The default weights are:

Category	Weight	Meaning
Moderate	40%	Common bad-weather events
Severe	35%	Significant river flooding
Extreme	20%	Rare, high-impact storms
Catastrophic	5%	Tail-risk scenarios

Increasing the catastrophic weight produces more tail-risk scenarios but reduces statistical coverage of common events. Equal weighting (all 25%) is useful for sensitivity analysis but misrepresents real-world frequency.

Each category also has a **base intensity distribution** (μ, σ) from which the raw intensity factor is sampled. For example, a "severe" storm draws its factor from $\mathcal{N}(1.8, 0.30)$. Increasing μ makes the category more intense on average; increasing σ widens the spread.

3.3 Sequence Formation: Isolated vs. Multi-Storm

Not every storm is a single event. A deep low-pressure system can spawn multiple frontal waves over several days. The `sequence_probability` parameter controls how likely each intensity category is to produce a multi-storm sequence rather than remaining isolated:

Category	P(sequence)
Minimal	5%
Baseline	15%
Moderate	30%
Severe	50%
Extreme	70%
Catastrophic	85%

The physical intuition is straightforward: more intense weather systems naturally produce multiple precipitation events. Increasing these probabilities makes compound flooding more prevalent in the catalogue.

When a sequence is formed, it takes one of three types:

Doublet Two storms separated by a medium gap.

Cluster Three to four storms with medium gaps.

Persistent Four to five storms with short gaps, maximising compound flooding.

The `sequence_type_weights` parameter controls the mix for each intensity category. Severe and extreme storms lean towards clusters and persistent patterns.

3.4 Intensity Correlation Within Sequences

When a sequence contains multiple storms, their intensities are not independent. The model captures *atmospheric memory*: if Storm 1 is intense, Storm 2 is likely intense as well.

Three parameters govern this correlation:

`first_storm_dominant_prob` **(default 0.30)** The probability that Storm 1 is the strongest in the sequence. When this is low, later storms tend to dominate, producing escalating patterns.

`correlation_prob` **(default 0.70)** The probability that each subsequent storm maintains or exceeds the intensity of its predecessor. Higher values produce stronger compounding.

`intensity_variation` **(default ±0.20)** The fractional envelope around the base intensity. Narrower variation produces tightly correlated sequences; wider variation allows larger swings between pulses.

Increasing `correlation_prob` towards 1.0 ensures that nearly every second storm exceeds the first—a worst-case compounding assumption. Decreasing it towards 0.5 makes successive storms effectively independent.

3.5 Duration Sampling

Each storm pulse has a duration drawn from a **triangular distribution** $(min, mode, max)$ specific to its intensity category:

Category	Min (h)	Mode (h)	Max (h)	Physical system
Minimal	2	6	12	Quick showers
Baseline	6	12	24	Brief rainstorms
Moderate	12	30	60	2.5-day fronts
Severe	24	48	96	4-day depressions
Extreme	36	72	144	6-day frontal systems
Catastrophic	48	120	240	10-day blocking patterns

Longer duration means more total precipitation for the same intensity. Increasing the maximum for severe storms allows rare multi-day events; decreasing the mode produces punchier, shorter pulses with less total rain.

3.6 Gap Sampling: Inter-Storm Drainage

Between storms in a sequence, the river drains. The amount of drainage depends on the gap length:

Gap Type	Min (h)	Mode (h)	Max (h)	Recovery
Short	6	18	36	30–60%
Medium	24	48	72	60–80%
Long	48	96	144	80–95%

Persistent sequences use **short** gaps, maximising the compound effect: Storm 2 arrives before the river has drained from Storm 1. Doublets and clusters use **medium** gaps, allowing partial recovery.

Shortening the mode of the "short" gap (e.g. from 18 h to 10 h) dramatically increases compound flooding severity because the river starts Storm 2 from a higher baseline. Lengthening gaps reduces compounding and produces sequences that behave more like independent storms.

4 Hydrograph Synthesis

Once a storm sequence defines precipitation timing and intensity, the hydrograph synthesis stage converts each pulse into a water-level time series at the gauge. This is where rainfall becomes river flow.

4.1 Gamma-Shaped Hydrograph Template

Each precipitation pulse generates a dimensionless hydrograph shaped by the gamma function with shape parameter α. The `hydro_alpha` parameter varies by sequence type:

Sequence Type	α	Shape
Isolated	0.3	Fast rise, long tail (flashy)
Doublet	0.3	Fast rise, long tail
Cluster	0.7	Balanced rise and fall
Persistent	1.0	Broad, symmetric peak

Small α produces a sharp peak followed by a slow recession—characteristic of flashy urban catchments. Large α produces a broad, sustained peak—characteristic of large rural catchments where water takes longer to concentrate.

Increasing α for isolated storms would flatten their peaks and extend their duration, reducing the maximum water level but keeping the river elevated for longer.

4.2 Antecedent Saturation

When multiple storms occur in sequence, the soil and groundwater are already partially saturated from the preceding event. The saturation model amplifies later peaks using the formula:

$$s_i = 1 + \beta \times \ln\left(1 + \frac{A_i}{P_0}\right)$$

where A_i is the accumulated precipitation from all prior pulses, β is the `saturation_beta` parameter (default 0.2), and P_0 is the reference precipitation `saturation_p0_mm` (default 50 mm).

Increasing β strengthens the amplification: a sequence that previously produced a 10% boost to Storm 2 might produce a 20% boost. This is the primary control for compound flooding intensity. Increasing P_0 reduces the saturation effect by normalising against a larger baseline.

4.3 Superposition and Capping

When multiple pulses overlap or follow closely, their hydrographs are linearly superimposed. Without a cap, mathematical superposition of five pulses could produce unrealistic water levels exceeding channel capacity. The `superposition_cap_factor` (default 2.5) limits the peak to:

$$\text{cap} = \text{base} + 2.5 \times (\text{severe} - \text{base})$$

Increasing the cap factor allows higher compound peaks. Decreasing it constrains the maximum water level, which may underestimate extreme compound events but prevents physically impossible readings.

4.4 Infiltration Along the Flow Path

As flood water moves from the river channel towards a property, some volume is absorbed into the ground. The infiltration model tracks cumulative absorbed depth:

`infiltration_rate_per_hour` **(default 0.005)** The hourly absorption rate. Higher values reduce flood depths at distant properties.

`infiltration_ymax_ref_m` **(default 0.10 m)** The maximum depth the ground can absorb (fully pervious). Once exhausted, no further infiltration occurs.

`default_imperv_fraction` **(default 0.4)** Fraction of impervious surface. Effective absorption capacity is $(1 - f_{\text{imperv}}) \times y_{\max}$. Urban corridors (high impervious fraction) lose less water to infiltration, producing deeper floods at properties.

4.5 Depth-Damage Vulnerability Curve

The final translation from flood depth to financial loss uses a piecewise-linear curve calibrated to UK flood claims data. The `depth_points` and `damage_points` arrays define the control points:

Depth (m)	Damage ratio
0.00	0%
0.05	5%
0.50	25%
1.00	40%
1.50	50%
2.00	60%
3.00	75%
4.00	85%
5.00	95%
6.00	100%

The curve is non-linear: shallow floods cause proportionally less damage (carpets and skirting boards), while deep floods destroy structural elements. Shifting the control points upward makes properties more vulnerable at each depth; shifting them downward makes properties more resilient.

5 Gauge-to-Property Propagation

A gauge measures water level at a single point on the river. Properties are scattered across the floodplain at varying distances and elevations. This stage translates gauge readings into property-specific flood depths.

5.1 Manning's Equation and Flow Velocity

Flood water spreads across the floodplain at a velocity governed by Manning's equation:

$$v = \frac{1}{n} R^{2/3} S^{1/2}$$

where n is the roughness coefficient (`default_roughness`, default 0.04), R is the hydraulic radius, and S is the terrain slope.

Increasing roughness slows the flow, delaying flood arrival at distant properties but not necessarily reducing peak depth. Decreasing roughness speeds flow, meaning water arrives sooner but also drains faster.

The `min_slope` parameter (default 0.001) prevents division instability on flat terrain where the actual slope is near zero.

5.2 Retention: Exponential Decay with Distance

As water moves away from the channel, peak depth decays exponentially:

$$\text{retention} = \exp\left(-\frac{d}{L}\right)$$

where d is the distance from the river and L is the `default_retention_length` (default 10,000 m). At $d = L$ the retention factor is $1/e \approx 0.37$; at $d = 0$ retention is 1.0 (full gauge signal).

Increasing the retention length extends the reach of flooding—properties further from the river receive a larger fraction of the gauge signal. Decreasing it confines flooding to a narrow corridor near the channel.

5.3 Terrain Velocity Scale

Different land cover produces different flow velocities. The `terrain_velocity_scale` parameter adjusts velocity relative to an urban baseline:

Terrain	Scale	Physical basis
Urban	1.0	Streets, hard surfaces ($n \approx 0.025$)
Semi-urban	0.7	Gardens, parks, mixed ($n \approx 0.035$)
Rural	0.5	Pasture, crops ($n \approx 0.050$)
Floodplain	0.3	Heavy vegetation, wetland ($n \approx 0.080$)

Lower velocity means slower flow and more attenuation on the floodplain. Properties in rural or floodplain terrain are reached later and experience lower peak depths than those in urban terrain at the same distance.

5.4 Bankfull Offset

Overbank flooding begins when water exceeds the **bankfull level**, not the severe warning threshold. The severe level is an administrative safety alert set above bankfull for UK rivers. The `bankfull_offset_m` parameter (default 0.5 m) defines this difference:

$$\text{bankfull} = \text{severe\_level} - \text{bankfull\_offset}$$

Increasing the offset means more water is needed before flooding begins, reducing the number of properties flooded from a given gauge reading. Decreasing it makes flooding trigger earlier.

5.5 Number of Nearest Gauges

The `n_nearest_gauges` parameter (default 3) controls how many gauges contribute to the flood estimate at each property via inverse-distance weighting (IDW). In practice, the nearest synthetic gauge dominates. Increasing this value adds spatial smoothing from more distant gauges.

5.6 Recession Factor

The `default_recession_factor` (default 1.5) makes drainage slower than arrival. A factor of 1.5 means the flood tail lasts 50% longer than the rising limb. Increasing it extends flood duration at properties; decreasing it towards 1.0 makes rise and fall symmetric.

6 Spatial Correlation

A single storm does not produce uniform rainfall across an entire catchment. The spatial correlation model distributes catchment-mean precipitation unevenly across gauges using an exponential kernel.

6.1 Exponential Correlation Kernel

The correlation between two gauges at distance d kilometres is:

$$C(d) = (1-\varepsilon)\exp\left(-\frac{d}{r}\right) + \varepsilon\,\delta_{ij}$$

where r is the effective range and ε is the `spatial_corr_nugget` (default 0.05). The nugget represents micro-scale variability that is uncorrelated even between co-located stations.

The range scales with storm intensity:

$$r = r_0 \times \big(1 + \rho \times (I-1)\big)$$

where r_0 is the `spatial_corr_base_range_km` (default 40 km, approximately half the Thames corridor) and ρ is the `spatial_corr_rho_intensity` (default 0.40).

Increasing the base range makes rainfall more uniform across the catchment—all gauges receive similar amounts. Decreasing it produces localised rainfall cells where one gauge may receive heavy rain while a neighbour 20 km away stays dry.

Increasing ρ makes extreme storms even more spatially coherent (wider footprint), while leaving moderate storms localised.

6.2 Lognormal Precipitation Field

The correlated random variates are transformed into a lognormal field with standard deviation `spatial_corr_sigma` (default 0.40), corresponding to approximately 40% coefficient of variation in rainfall across the catchment.

Increasing sigma produces more spatial contrast: some gauges receive double the catchment mean while others receive half. Decreasing it towards zero makes rainfall nearly uniform (all gauges get the same amount).

The `spatial_corr_enabled` toggle allows the spatial model to be switched off entirely, in which case every gauge receives the catchment-mean precipitation—useful for debugging or testing in a simplified mode.

7 Stress Catalogue

After all 20,000 sequences are generated and propagated, the system filters them to build a stress testing catalogue.

`stress_storms_min_count` **(default 50)** Minimum number of alert-breaching storms required per gauge. Gauges with fewer qualifying storms are excluded from stress testing. Increasing this threshold requires more storms to pass the filter, potentially excluding gauges with low flood frequency.

`stress_storm_default_duration_hours` **(default 168)** Fallback hydrograph duration used when storm metadata does not include explicit timing. Should match the event window.

stress_storm_default_peak_position (default 0.5) Fractional position of the peak within the storm window. A value of 0.5 centres the peak; 0.3 produces an early peak with a long tail; 0.7 produces a late-arriving peak.

8 Parameter Reference

The following tables list every parameter on the Control screen with its default value and the effect of adjustment.

8.1 Storm Generation Parameters

Parameter	Default	Increase	Decrease
`event_window_hours`	168	Longer storm sequences possible	Sequences must fit in shorter window
`min_drainage_window`	12 h	More guaranteed recession	Precipitation can extend later
`intensity_variation`	±0.20	Wider intensity swings between pulses	Tighter correlation to base
`first_storm_dominant`	0.30	First storm more likely strongest	Later storms tend to dominate
`correlation_prob`	0.70	Stronger compounding (Storm 2 ≥ Storm 1)	Storms more independent
`default_type_weights`	[0.6, 0.3, 0.1]	More doublets if first weight rises	More persistent if third rises
`sequence_probability`	0.05–0.85	More multi-storm sequences	More isolated storms
`intensity_weights`	see text	Heavier tail if catastrophic rises	More common events dominate
`catchment_base_precip`	35–55 mm	More rainfall per unit intensity	Less rainfall per unit intensity
`duration_params`	see text	Longer storms, more total rain	Shorter, punchier storms
`gap_params`	see text	More drainage between storms	Less drainage, stronger compounding
`base_intensity_params`	see text	Higher mean intensity per category	Lower mean intensity

Parameter	Default	Increase	Decrease
`sequence_type_weights`	see text	Shift toward persistent patterns	Shift toward doublets

8.2 Hydrograph Synthesis Parameters

Parameter	Default	Increase	Decrease
`hydro_alpha`	0.3–1.0	Broader, flatter peaks	Sharper, flashier peaks
`saturation_beta`	0.2	Stronger compound amplification	Weaker saturation effect
`saturation_p0_mm`	50 mm	Reduces saturation sensitivity	Amplifies saturation effect
`infiltration_rate`	0.005/h	More absorption, lower flood depths	Less absorption, higher floods
`infiltration_ymax`	0.10 m	Ground absorbs more before saturating	Ground saturates sooner
`imperv_fraction`	0.4	Less infiltration capacity (urban)	More infiltration (rural)
`superposition_cap`	2.5×	Allows higher compound peaks	Constrains peak levels
`depth_points`	see text	Stretches damage curve horizontally	Compresses damage curve
`damage_points`	see text	More damage at each depth	Less damage at each depth

8.3 Gauge-to-Property Propagation Parameters

Parameter	Default	Increase	Decrease
`default_roughness`	0.04	Slower flow, more attenuation	Faster flow, less attenuation
`retention_length`	10,000 m	Flooding extends further from river	Flooding confined near channel
`min_slope`	0.001	Higher minimum velocity on flat terrain	Lower floor, potential instability

Parameter	Default	Increase	Decrease
recession_factor	1.5	Longer flood duration	Faster drainage (symmetric rise/fall)
bankfull_offset_m	0.5 m	More water needed to trigger flooding	Flooding begins earlier
n_nearest_gauges	3	More spatial smoothing	Nearest gauge dominates
terrain_velocity	see text	Faster flow for that terrain type	Slower flow, more attenuation

8.4 Spatial Correlation Parameters

Parameter	Default	Increase	Decrease
spatial_corr_enabled	true	N/A	Disables spatial model (uniform rain)
base_range_km	40 km	More uniform rainfall across catchment	More localised rainfall cells
nugget	0.05	More micro-scale variability	Smoother spatial field
rho_intensity	0.40	Extreme storms more spatially coherent	Range independent of intensity
sigma_lognormal	0.40	More spatial contrast between gauges	More uniform rainfall

8.5 Stress Catalogue Parameters

Parameter	Default	Increase	Decrease
stress_storms_min_count	50	Excludes gauges with few qualifying storms	Includes more gauges
default_duration_hours	168 h	Longer fallback hydrograph	Shorter fallback
default_peak_position	0.5	Later peak (delayed climax)	Earlier peak (front-loaded)

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